Title

Your Name

6. Mai 2026

Abstract

Abstract text.

Zusammenfassung

Abstract text.

Nomenclature

Physical nomenclature:

- SKA: Square Kilometer Array
- EoR: Epoch of Reionization
- SM: Standard Model
- WDM: Warm Dark Matter
- ΛCDM: Lambda Cold Dark Matter
- IGM: Intergalactic Medium
- FRW: Friedmann-Robertson-Walker

Mathematical/statistical nomenclature:

- SBI: Simulation-Based Inference
- CNN: Convolutional Neural Network
- MLP: Multilayer Perceptron
- KL: Kullback–Leibler divergence
- CLT: Central Limit Theorem
- MI: Mutual Information
- CE: Conditional Entropy

Notation

Standard probability theory notation is adopted. In particular, the following notations are used:

- X for random variables with sample space $\mathcal{X}$
- $\mathcal{P}(\mathcal{X})$ for the set of all probability measures on the Borel σ-algebra over $\mathcal{X}$
- $P \in \mathcal{P}(\mathcal{X})$ for probability measures
- p_X for the density function of a probability measure P, index might be dropped if notation is clear
- $X \sim p$ for a random variable X whose distribution is described by p

1 Introduction

I am currently still fighting with the citations, that will hopefully soon be resolved.

The current cosmological standard model predicts the epoch of Cosmic Dawn (CD) and the subsequent Epoch of Reionization (EoR) as mechanisms for star and structure formation and the following ionization of the intergalactic medium (IGM) at a redshift range of roughly $z \sim [5-20]$. A number of sources from multi-messenger astronomer have contributed to this current model: Scattering of CMB photons on free electrons during and after reionization can be detected once IGM became ionized significantly enough. This places the beginning of EoR at $z \sim 8$, though with a large uncertainty Lymann-alpha forests in the spectra of distant quasars and galaxies are used to estimate the neutral hydrogen fraction at low redshifts, placing the end of the EoR at around add here citation. The JWST with an observational range of $z \sim$ ... measures earliest galaxies and will be able to additionally constrain the speed of galaxy evolution.

A comparatively new probe is that of the 21 cm hyperfine transition line of neutral hydrogen, which constitutes a background spectrum similar to that of the CMB. It is the only known probe with which the Dark Ages could be studied and also the one to study CD to the highest redshifts. First experiments have measured reionization profiles in terms of power spectra and signals of possible dark matter presence cite. Much better results are expected from the Square Kilometer Array (SKA) Observatory, which is a radio telescope currently in construction which will image the 21 cm line of the IGM at a redshift range of $z \sim [6-30]$ with unprecedented accuracy. Hopes are that this signal can be used to significantly improve the current understanding of the reionization history and structure formation and possibly act as a new dark matter probe.

There are multiple difficulties in interpreting the 21 cm signal. Traditional Gaussian summary statistics do not capture enough of the signal's structure, reflecting the fact that the processes being observed are highly nonlinear. This also entails that the likelihood of the 21 cm signal is intractible. Additionally, the foreground contamination for ground-based observatories such as the SKA will be significant.

These challenges of non-Gaussianity and foreground noise motivate the use of neural networks for the analysis of SKA data. Another sentence on NNs here. Neural networks are nowadays commonly used in Bayesian inference, which attempts to reconcile prior knowledge and empirical data in order to find optimal parameters describing, for example, our universe.

The issue of an intractible likelihood is tackled with so-called simulation-based inference. The idea behind SBI is to train a NN on simulated data, were the inference parameters are known. Once the model is trained, inference is amortized: the model can perform inference on real data with the pre-trained model inexpensively. The prerequisite is, of course, that the model generalizes well from the training data to the real data. This approach has successfully bin implemented in ergänzen.

Not all models are equally good for a given problem and as a consequence, the choice of model introduces bias to the inference. In an attempt to find a principled way to choose a model, one can use methods developed in information theory, which can here be used to quantify the quality of a model.

In this thesis, a model typically used in parameter inference for simulated SKA-data will be analyzed with the methods of information theory.

2 Physics Background

A few topics: - short introduction to history of universe - then focus on EoR - 21 cm signal from QM and how it can be used to learn about the EoR

Still need some more subsections to explain the other parameters being examined

2.1 The Cosmological Setting

For the purposes of this thesis, a flat ΛCDM cosmology with an additional warm dark matter parameter is assumed. The ΛCDM-model is based on the Friedmann-Robertson-

Walker metric:

$$ds^2 = -dt^2 + a^2(t)\left(\frac{dr^2}{1-kr^2} + r^2\left(d\theta^2 + \sin^2\theta d\phi^2\right)\right). \tag{1}$$

where $a(t)$ is the dimensionless scale factor and k is the curvature constant.

Combined with Einstein's field equations and the assumption that the universe may be described as a perfect fluid, the dynamics of this universe are described by the Friedmann equations:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3} - \frac{k}{a^2}, \tag{2}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p) + \frac{\Lambda}{3}. \tag{3}$$

where G is Newton's constant, Λ is the cosmological constant and ρ is the energy density of the universe. The scale factor as well as the energy density are time-dependent quantities.

The ratio $H = \dot{a}/a$ is called the Hubble parameter and measures how fast the universe is expanding in the following sense: given two observers at $(t_1, r_1), (t_2, r_2)$ in a FRW-universe, where observer 1 emits a signal of duration Δt_1, observer 2 seeds a scale-factor dependent dilation of the signal duration, as measured by the cosmological redshift:

$$z = \frac{\Delta t_2}{\Delta t_2} - 1 = \frac{a(t_2) - a(t_1)}{a(t_1)} > 0 \quad \text{if universe expanding} \tag{4}$$

In the limit of $t_2 \to t_1$, if d is the instantaneous distance between the observers, one then has

$$z \approx \left.\frac{\dot{a}}{a}\right|_{t_1} d + \mathcal{O}((t_1 - t_2)^2) \tag{5}$$

such that H measures the expansion rate at a given time in units 1/s.

Now a part on matter densities

The six parameters describing the ΛCDM-model and their estimated values today are shown in table 1.

See PieNET paper:

All other cosmological parameters are fixed to the Planck measurements, assuming flatness and a cosmological constant. This means Ωb $= 0.04897$, $\sigma_8 = 0.8102$, h$= 0.6766$, and ns $= 0.9665$.

These are not the fundamental params. Choose which exactly to include.

Table 1: Parameters of ΛCDM with estimated values

Name	Symbole	Value	Unit	Source
Hubble constant	H_0	67.4	$\mathrm{km\,s^{-1}\,Mpc^{-1}}$	Planck 2018
Matter density	Ω_m	0.315 ± 0.007		Planck 2018
Dark energy density	Ω_Λ	0.685		Planck 2018
Spectral index	n_s	0.965		Planck 2018
Matter fluctuation amplitude	σ_8	0.811 ± 0.006		Planck 2018
Equation of state	w	-1		Planck 2018

Cite a few papers that give the current status of cosmology in ΛCDM, e.g. Planck 2018, etc.

Cite current values for all params that will be inferred

According to the cosmological standard model, the timeline of the universe is approximately as follows:

The first phase of the universe is marked by *radiation domination*. Immediately after the Big Bang, the universe underwent a period of rapid expansion called *inflation*, during which space expanded by appriximately an order of 10^{26}. This inflation was driven by the so-called inflaton field. During the following process of reheating, the energy of the inflaton field was converted to SM particles, which enabled subsequent *baryogenesis*, i.e. the creation of matter and antimatter particles. In the subsequent annihilation of matter.antimatter pairs, a small excess of matter paticles survived, composing the matter content of the universe today. There have been many proposed mechaninisms for both reheating and baryogensesis, and both are active fields of research. Insert sources here In the following time scale of $20 \cdot 10^{-12}\mathrm{s} - 1000\mathrm{s}$, particles acquired mass and coalesced into hadrons. Neurinos decoupled from baryons and thus the cosmic neutrino background was formed. What followed was an epoch named *big bang nuceosynthesis*, during which mostly protons and neutrons bound to primordial nuclei, consisting mostly of hydrogen, Helium-4 and traces of deuteron, Helium-3 and lithium.

The second phase of the universe is marked by *matter domination*. During *recombina-*

tion, electrons and nuclei where bound to neutral atoms. Photons were no longer in thermal equilibrium with matter and thus "decouple", forming the cosmic microwave background (CMB). Measurements of the CMB today provide one of the most crucial constraints on cosmological parameters.

After recombinations, most photons propagate freely and are quickly redshifted, such that the universe becomes dark. This period is therefore called the *dark ages*. Matter has not yet coalesced into stars and galaxies. The main source of photons visible phtons? is the 21 cm emission line of neutral hydrogen.

Once first stars and galaxies have formed, they emit photons energetic enough to ionize the surrounding hydrogen. This period is called the *epoch of reionization* (EoR). By the end of the EoR, most of the IGM is ionized and continues to be so today.

Need small intro on warm dark matter

These sections are based on

Maybe short example of how an evolution might be described...

How other cosmological models can be tested with the EorFlow setup, maybe also put this somewhere else.

What is matter distribution at the redshift when beginning study?

Brief summary of CMB

Small overview of existing studies of timeline of universe, why there is a gap in EoR

Maybe cite more recent paper that explains why studying this exact regime of redshifts

2.2 The Astrophysical Setting

2.3 On the Origin of the 21 cm Signal

Temperature of the ISM at around these redshifts, why neutral hydrogen does not absorb of emit visible light at these temperatures.

How trace elements interact with visible light at these temperatures.

At least mention the possibility of a derivation

Quantum mechanical aspect: hyperfine splitting of ground state of hydrogen.

Statistical mehanics aspect: spin temperature and how it is useful in measuring the 21 cm signal.

- Einstein coefficients from QM - Different processes that influence the spin temperature, e.g. collisional coupling, Wouthuysen-Field effect, etc. - Final formula for spin temperature

2.4 Evolution of the 21 cm Signal

3 Simulations

Will have a part on SBI here

Could either put standard cosmology here or into previous section. Need to explain params used in simulations.

21 cm fast

The parameters used, and the priors on them. Why these parameters? What do they mean physically?

The six parameters and their priors chosen for the simulations in this project are summarized in table 2. The priors are chosen to be uniform and constrained to a range motivated by multimessenger observations.

Table 2: Parameters used for the simulations

Name	Symbol	Prior
Matter density	Ω_m	$\mathcal{U}(0.2, 0.4)$
Warm dark matter mass	ω_{WDM}	$\mathcal{U}(0.3, 10)$ keV
Minimum virial temperature	T_{vir}	$\mathcal{U}(10^4, 10^{5.3})$ K
Ionization efficiency	ζ	$\mathcal{U}(10, 250)$
Specific X-ray luminosity	σ_8	$\mathcal{U}(10^{38}, 10^{42})$ erg/s/M$_\odot$ $\times$ yr
X-ray energy threshold for self-absorption by host galaxies	E_0	$\mathcal{U}(100, 1500)$ eV

interpretation of data

Noise simulations

If in end used different types of data, then explain them here. And why less params where used for information geometry.

How reioniazation history is calculated

Need to clarify which physical quantities are being simulated, i.e. temp. and axes are frequency and angular coords.

4 Information Theory and Neural Estimators

Information theory: - Entropy, KL divergence, mutual information - Interpretation of these quantities, maybe expectation?

Information geometry: - Fisher information metric, geodesic distance, curvature - Interpretation of these quantities, maybe expectation?

4.1 Some Basic Measures in Information Theory

Given a random variable $X \sim p$, the *entropy* of X is defined as:

$$\mathbb{H}(X) = -\mathbb{E}_X \left[\log p(X)\right] \geq 0. \tag{6}$$

If the logarithm is chosen as with base 2, one defines the unit of this quantity as bits. If one uses the natural logarithm, the entropy is measured in units of nats.

Entropy is a measure of the uncertainty of a random variable. A high entropy corresponds to a lack of predictability of the random variable's outcome. As an example, the entropy of the uniform distribution is maximal, while that of the delta distribution is zero. One therefore considers entropy to be a measure of the amount of information contained in the random variable's distribution.

The generalization to conditional entropy is:

$$\mathbb{H}(X|Y) = -\mathbb{E}_{X,Y} \left[\log p(X|Y)\right] \tag{7}$$

Another important metric in information theory is the *Kullback-Leibler divergence*. Let $X \sim p$ and $Y \sim q$ be two random variables over the same sample space. The

KL-divergence between p and q is defined as:

$$\mathbb{D}_{\mathrm{KL}}(p||q) = \mathbb{E}_X \left[\log p(X) - \log q(X)\right] \geq 0 \tag{8}$$

The KL-divergence satisfies the identity axiom, $\mathbb{D}_{\mathrm{KL}}(p||q) = 0 \Leftrightarrow p = q$. It is not symmetric and does not satisfy the triangle inequality and as such is not a metric. However, it does still carry an interpretation as a distance between two probability distributions: from the above expression it is clear that the more q resembles p, the smaller the measured KL-divergence becomes. A smaller the KL-divergence between the two distributions means there is less information loss by using q to approximate p.

Maybe a word about consequece of assymetry of D KL

Could also talk about why mionimizing D KL is often equivalent to common optimization objectives.

Lastly, the *mutual information* between two random variables X and Y is defined as:

$$\mathbb{I}(X;Y) = \mathbb{D}_{\mathrm{KL}}(p(X,Y)||p_X(X)p_Y(Y)) \geq 0. \tag{9}$$

An alternative formulation in terms of entropy is:

$$\mathbb{I}(X;Y) = \mathbb{H}(X) - \mathbb{H}(X|Y) = \mathbb{H}(Y) - \mathbb{H}(Y|X). \tag{10}$$

Mutual information is symmetric and nonnegative. It is zero iff X and Y are independent and becomes maximal iff $X = Y$. One interprets it's value as the amount of information obtained about one random variable while observing another.

4.2 Mutual Information Estimation

In the context of representation learning, mutual information can be used as a measure of model quality. This is motivated by the *data processing inequality*, which in this context can be phrased as follows:

Let X, Y and Z be random variables where X is the original data, Y is a learned summary of X and Z is a label associated with X. These form a Markov chain $X \to Y \to Z$, where Z is only inferred from Y. Then Z cannot contain more information about X than Y, so

$$\mathbb{I}(X;Y) \geq \mathbb{I}(X;Z) \tag{11}$$

A good representation keeps as much information about the original data as possible, subject to the constraint that it should contain less total information than the data itself, i.e. summarize it in some way. Mit Formel unterstützen

It also captures more information than just covariance!

Something on summary statistics?

In practice, this necessitates an estimation of mutual information between the representation and the target distribution.

In the setting of variational inference, one can use the learned posterior to estimate mutual information in a natural way, described in the following:

$$\mathbb{I}(X;Y) = \int_{\mathcal{X}\times\mathcal{Y}} p(x,y) \log \frac{p(x|y)}{p(x)} \ , \tag{12}$$

where the distribution $p(x|y)$ is estimated by a learned $q_\theta(x|y)$, the optimization objective being to minimize the KL-divergence $\mathbb{D}_{\mathrm{KL}}(p||q_\theta)$. The Monte-Carlo approximation of equation (12) using this distribution is referrred to as the *Barber-Agakov estimate*

$$\hat{\mathbb{I}}(X;Y) = \mathbb{I}_{\mathrm{BA}}(X;Y) = \mathbb{E}_{P(X,Y)}\left[\log q_\theta(x|y) - \log p(x)\right] \tag{13}$$

This is computable, because the distribution of the prior $p(x)$ can in practice either be ignored as a constant factor, or in case of SBI is chosen and therefore known. If this is not the case, $p(x)$ can also be estimated separately, which introduces more uncertainty in the above estimate.

Because the KL-divergence is non-zero, the BA-estimate yields a lower bound on the true mutual information:

$$\mathbb{I}(X;Y) \geq \mathbb{I}_{\mathrm{BA}}(X;Y) \tag{14}$$

Maybe something about the bias of the estimator oder so...

Why estimating MI in high dimensions is hard, and quantify!

Also another method of approximation, maybe Donsker-Varadahn (MINE) ?

Maybe something on limitations on the interpretation of these metrics. Could also tease limitations in high dimensions, or give intuition for amount of data needed.

5 Basic Neural Networks

EoRFlow general description, goal etc. Why it is needed. Have figure of architecture.

How this type of model can be used in 21 cm cosmology in general, and then how it is used here specifically.

Traditionally, data as complex as those expected from SKA are analyzed using summary statistics. A summary statistic is a function that maps the data to a lower-dimensional space, typically with some physical interpretation. An example is the power spectrum. The interpretation of the power spectrum is blablabla. However the power spectrum is not a sufficient statistic for the 21 cm signal. As an alternative, one can use a neural network to learn a summary statistic of the data. While being potentially more informative, this comes at the cost of less interpretability.

For cnn and flow: - general stuff on model - architecture, where it's inspired from, how it was adapted to the problem, etc - loss function - backpropagation

Training - details on scheduler, optimizer, etc - training hyperparameters, chip used

Why is a generative model needed? Why is the flow chosen? What are the advantages of flows over other generative models?

Why this very specific architecture?

Do I want a short ML intro? Could e.g. introduce MLP here and rename section to NNs in general

A word about how architectures will be discussed, e.g. notation with h, z, x, etc.

5.1 Convolutional Neural Networks

CNNs are inspired by the task of implementing neural networks in image analysis, i.e. on data with 2D spatial structure. An early version was successfully implemented in 1983 with Fukushima's "Neucognitron" Neocognitron citation. An architecture implementing backpropagation was LeCun's "LeNet" LeNet citation.

To explain the novel concepts behind the CNN in comparison with a MLP, the CNN architecture will first be discussed in 1D. The generalization to higher dimensions will be discussed afterwards.

Let X denote the data space and Y the feature space.

Let the input have dimensions $x \in \mathbb{R}^{C \times H \times W}$ where C the channel dimension, H the height dimension and W the width dimension. In case of an RGB image, this channel dimension would be three.

The CNN learns a function $f_\theta : X \to Y$.

As opposed to a simple MLP, the CNN offers the following advantages:

Using summation convention

Not writing bias explicitly

A basic MLP consists of only one type of layer. If h is the output of the previous layer, the current layer is defined by

$$z_i = \phi(W_{ij} h_j), \tag{15}$$

The CNN implements additional layers.

A convolutional layer applies a convolution operaton:

Talk abt convolution vs cross-correlation here

Is this def even correct? Not need different indices on h?

$$z_i = \phi([w \circledast h]_i) = \phi(\sum_k w_k h_{i+k}) \tag{16}$$

where the dimensionality of the kernel $w \in \mathbb{R}^K$ is typically much lower than that of the data.

A pooling layer applies a pooling operation:

$$z_i = \operatorname{pool} K_{ij} h_j \tag{17}$$

where K defines a pooling window for each index i. Typical choices for pool are maxpool, which chooses the maximum of all given values, or meanpool, which calculates their mean.

The purpose of a pooling layer is to...

Add normalization layers here

Add info on stride params etc here

Put it together here, potentially add image of total architecture.

5.2 Normalizing Flows

Choose a Schreibweise. Use p_u if necessary, otherwise $u \sim p(u)$?

Differentiate the true target distribution and the modelled one. Then going for Kullback Leibler of the two is equivalent to maximizing log likelihood.

A normalizing flow model is a generative model that learns to transform a simple (often Gaussian) distribution into a more complex one. The transformation is a composition of invertible functions, which are chosen to be computationally efficient to invert.

Let $u \sim p_u = \mathcal{N}(0, I_n)$ with $n \in \mathbb{N}$ the base distribution and $x \sim p_x$ how to signal that x is n-dimensional? the target distribution.

The normalizing flow learns a transformation given by an invertible function $f : \mathbb{R}^n \to \mathbb{R}^n$, such that sampling from the base distribion and applying the transfomation yields samples from the target distribution.

Let $f : \mathbb{R}^n \to \mathbb{R}^n$ a differentiable, invertible function with $g = f^{-1}$ its inverse (i.e. f is a diffeomorphism). Then the model is a normalizing flow if:

$$x = f(u) \sim p_x, \quad u \sim p_u. \tag{18}$$

Should have two sim signs, no? How to differentiate sampling and being part of distribution?

Once the flow is learned, the process to generate samples from p_x is $u \sim p_u$ and subsequent $x = f(u)$.

In order to compute $p_x(x)$ for a given x, on applies the change of variables formula:

$$p_x(x) = p_u(g(x)) \left|\det\left(\frac{\partial g(x)}{\partial x}\right)\right| = p_u(u) \left|\det\left(\frac{\partial f(u)}{\partial u}\right)\right|^{-1}. \tag{19}$$

The normalizing flow becomes expressive by allowing f to be as complex as possible, while restricting f to a certain class of functions with efficiently computable Jacobian determinants makes it computationally tractable.

Come up with nicer notation for $g_i(\cdot|\theta_i)$

In practice, one often models g as a composition of simple neural networks, e.g. multilayer perceptrons. Let $g_\theta = g_1 \circ g_2 \circ ... \circ g_k$ with $g_i : \mathbb{R}^n \to \mathbb{R}^n, x \mapsto g_i(x|\theta_i)$ and $k \in \mathbb{N}$.

Now the loss becomes

$$\mathrm{NLL}_\theta = \sum_{i=1}^{k} \log|\det J(g_i)| - \log p_u(g_\theta(x)) \tag{20}$$

Now what this specific Freia flow does. Research on website!

Or maybe put exact architecture in appendix?

5.3 Variational Autoencoders

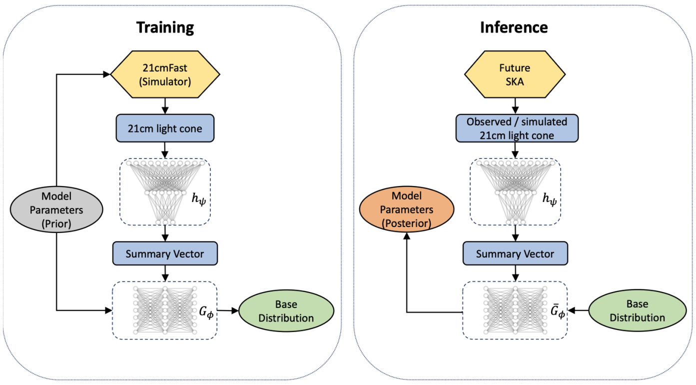
6 Specific Architectures

The following setup is common in current 21cm SBI:

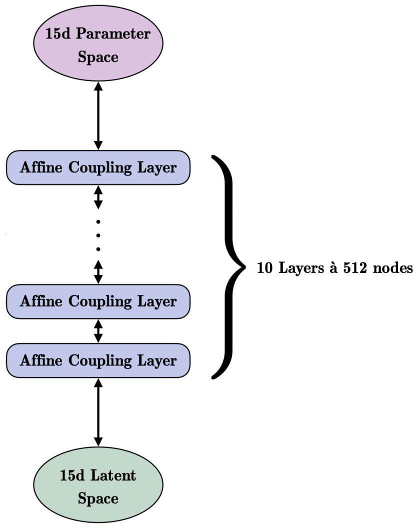
One networks learns a compression of the image data to a much lower-dimensional latent space, i.e. a summary statistic. Another network learns a mapping from the latent space to the parameter space.

The compressing network usually has a much heavier architecture. Once it has been trained, one can learn different lighter networks to map its summary to different parameter spaces. Example for parameter spaces include the parameters used for the simulations, or a simple physical summary statistic of the image data.

Maybe put some more examples, e.g. SKATR?



(a) Architecture of 21cmPIE-INN, taken from citation



(b) Architecture of the flow used in EoRFlow, taken from citation

Figure 1: Overall caption for both figures

Which construction is PIE-INN based on? Is it completely new?

One such construction is *21cmPIE-INN*, which consists of a CNN acting as a compressing network and a normalizing flow mapping its output to six parameters used for the original simulations. The architecture is visualized in figure 1a.

It is also possible to learn e.g. the reionization history with variable resolution from the CNN's summary, as is done with the *EoRFlow* architecture Citation here. The flow implemented in EoRFlow is slightly more complex the the 21cmPIE-INN. It's architecture is visualized in figure 1b.

7 Implementation

Maybe put choice of priors here actually. Then say that label distribution is not uniform.

The dataset consists of 5000 lightcones simulated with 21cmFast, cite the paper somewhere! and the calculated neutral hydrogen fraction as labels. It is split into a training, validation and test set with relative sizes 0.7, 0.2 and 0.1, respectively. Each lightcone is originally $140 \times 140 \times 2350$ voxels in size, corresponding to a redshift range of $z \sim [5, 12]$ find out spacial dimensions. In preprocessing, the temperature fluctuation values are normalized individually for each lightcone why is this better? and the lightcone is truncated to $140 \times 140 \times 1916$ voxels. The labels are originially 92-dimensional and are first truncated to 75 dimensions and then binned to 15-dimensional vectors. This step reduces the dimensionality of the inference problem. An additional logit transformation is applied to the labels, which improves training performance of the flow. A sigmoid transform can be applied during inference to map the labels back to their physical state.

In order to model the conditional distribution $p(x|y)$, the following steps are taken: An ensemble of models is trained for each bottleneck dimension. Each ensemble consists of multiple training runs at different weight initializations, i.e. different seeds. The CNN and flow are trained jointly. Both models were trained using an optimizer, a scheduler and a joint scaler. The optimization method is AdamW, the scheduler method is ReduceLROnPlateau and torch's GradScaler ensures more stable backpropagation. A summary of all the hyperparameters and their chosen values can be found in Cite Appendix. Maybe talk about Optuna?. Training is performed on one NVIDIA A30 GPU.

Unlike the original distribution of simulation parameters, the distribution of reionization history vectors is not uniform. It is therefore modelled separately. It is also modelled with a flow of the same architecture as the one from the conditional distribution, without any conditional input. This choice of model a priori adds additional uncertainty to the mutual information estimation. The idea behind choosing the same architecture to model the label distribution is that an artefact of the flow $q_\theta(x|y)$ leading to high variance in the MI estimation might be learned by $q_{\tilde{\theta}}(x)$ as well, thus leading to smaller variance in the final MI estimation.

8 Results

Does MI increase with accuracy on validation set? Does it increase with training time?

The average runtime for training is insert average runtime. Due to these training times, only specific bottleneck dimensions were analyzed. Bottleneck dimensions $1-3$ are analyzed for an estimate of lowest mutual information achievable with this model. Dimension 6 is analyzed because it equals the amount of parameters used to simulate the original lightcones. Dimensions $14-16$ are analyzed because one naively expects the model to be able to compress the information of 15-dimensional labels in a latent space of higher dimension. Given noise and a generally hard learning task, one might need more dimensions though, which is why dimensions 25 and 30 are also analyzed.

Updates on this to come, more models will be trained!

Two sources of error in the MI estimation will be discussed. The error from one training run, stemming from the variability of log probability values on the test set, will be referred to as the *model error*. The error of the MI estimation stemming from different weight initializations, i.e. seeds, will be referrred to as the *seed error*.

Figures [2–5] show the training and validation losses for each ensemble the model was trained on. Vertical lines indicate the epoch with best validation performance. Training stops after 250 epochs.

Comments on training here

One observes a higher seed variance for models with smaller bottleneck dimensions. There is also a higher variability in their training and validation losses across seeds. This indicates a higher training instability for models with smaller bottleneck dimension. These results are summarized in table 3Ist noch quatsch, richtige Werte folgen.

Table 3: Results on training stability

Bottleneck	Best epoch	Val loss	No. samples
4	38.81 ± 0.71	0.39	3
6	38.49 ± 0.95	0.83	4
15	39.90 ± 0.42	0.13	5
16	39.57 ± 0.46	0.13	4

Figures [6–9] show the estimated conditional entropy per epoch for each ensemble the model was trained on. Vertical lines again mark the epoch with best validation performance. All models reach a saturated estimate of the conditional entropy in the epoch range... The conditional entropy curves showcase a higher instability than the validation curves, which is attributed to the smaller size of the test set.

The model error on the conditional entropy estimate grows with increasing training steps. Some reionization histories are attributed with very low likelihood, such that the distribution of $\log q_\theta(\text{label}_i|\text{summary}_i)$ has a heavy left-sided tail. This is in parts

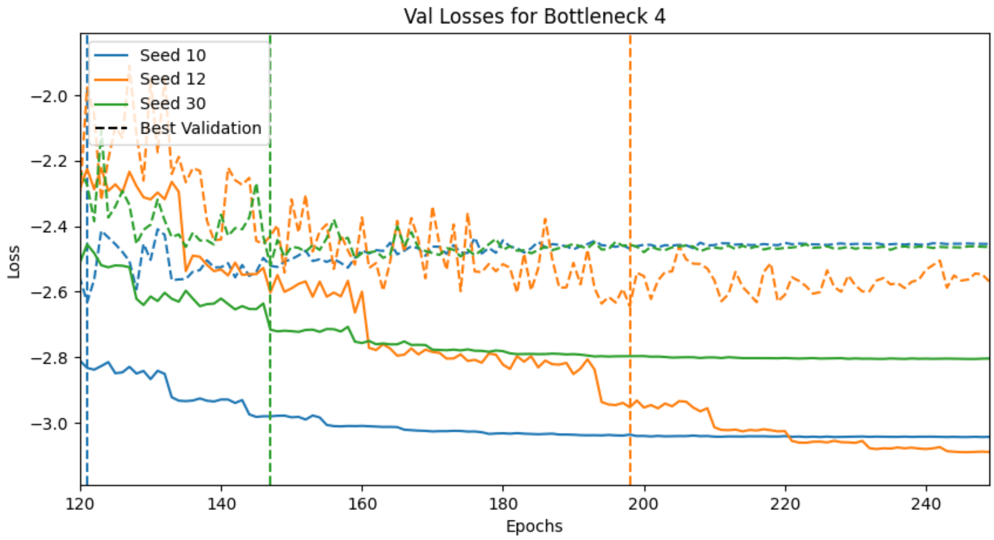


Figure 2: Train and Validation Losses for Bottleneck Dimension 4

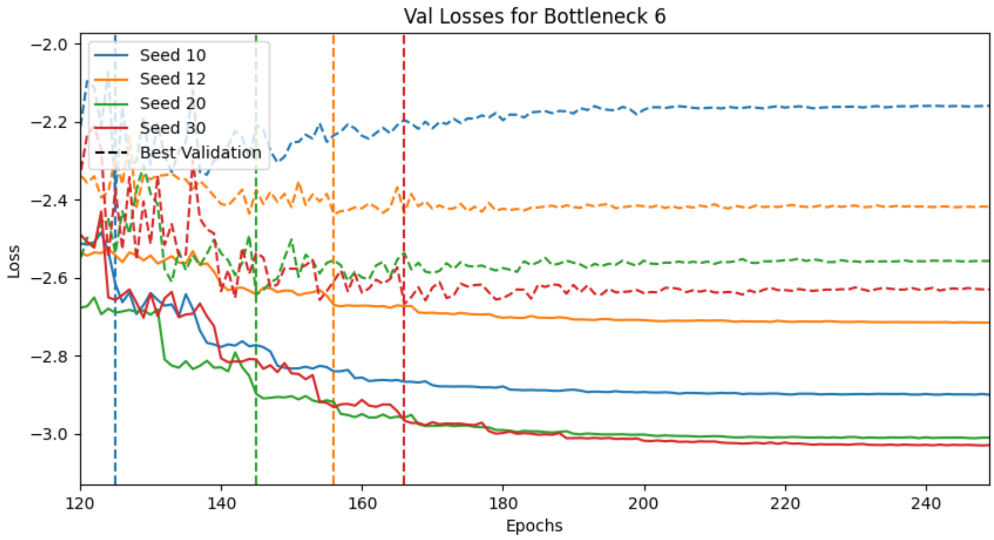


Figure 3: Train and Validation Losses for Bottleneck Dimension 4

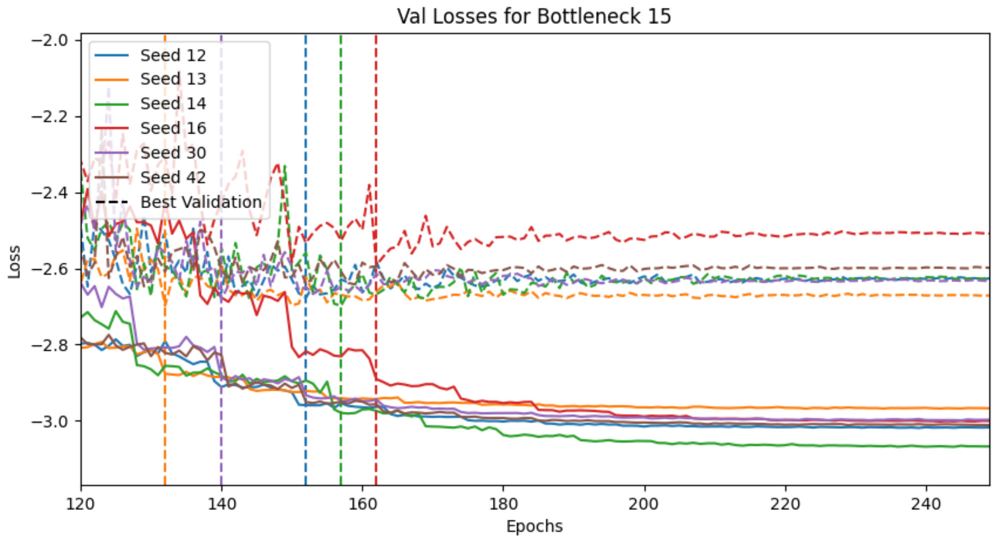


Figure 4: Train and Validation Losses for Bottleneck Dimension 4

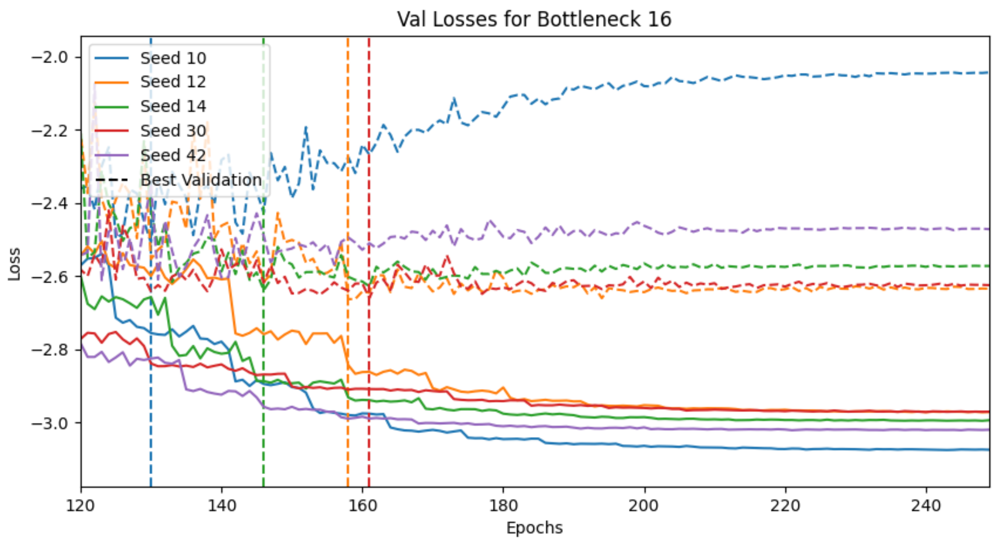


Figure 5: Train and Validation Losses for Bottleneck Dimension 4

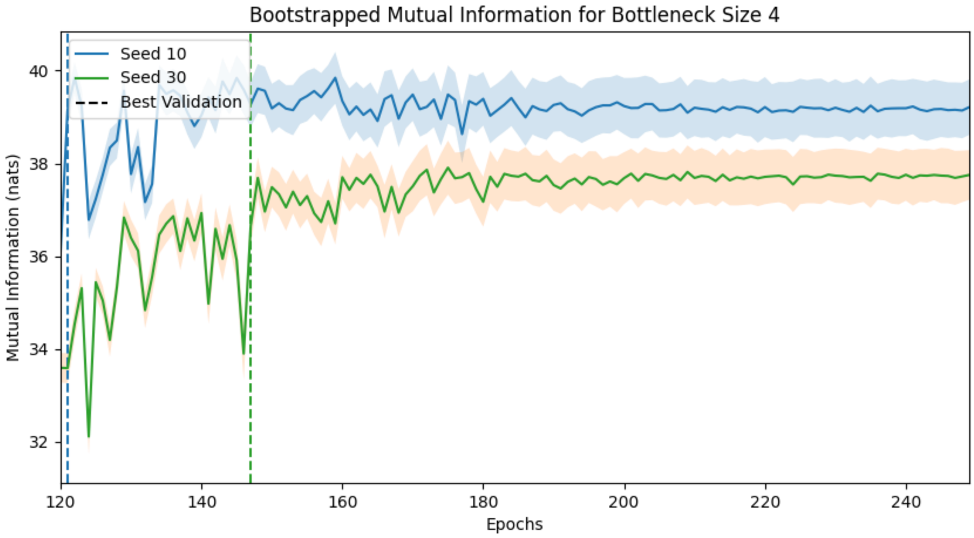


Figure 6: Mutual Information per Epoch for Bottleneck Dimension 4

because of the distribution of the labels themselves.

A separate normalizing flow is trained on the label distribution. Discuss other possible modelling choices and why not took them The model with best validation performance is chosen for the following analysis. It's distribution of $\log \tilde{q}_\theta(\text{label}_i)$ also showcases a heavy left-sided tail.

As an example, a histogram plot of the distributions $\log q_\theta(\text{label}_i|\text{summary}_i)$ and $\log \tilde{q}_\theta(\text{label}_i)$ is shown in figure insert figure.

According to the CLT, one would estimate the model error by the standard deviation. The heavy kurtosis inflates the value of standard deviation. As an alternative estimate of error, a bootstrapping method is implemented. Quote some source that cites this as best practice. The exact values used for bootstrapping for reproducibility. Something on interpretation of bootstrap results.

For each bottleneck dimension, the final MI estimate is made by selecting the model with best validation loss from each seed and

averaging the conditional entropy for each of these models. The final error on the CE estimate is comprised of the model error and the seed error, stemming from the variability of results across different seeds. The final estimates with errors as shown in table 4Noch sind das conditional-entropy werte, die aktuellen Werte folgen gleich.

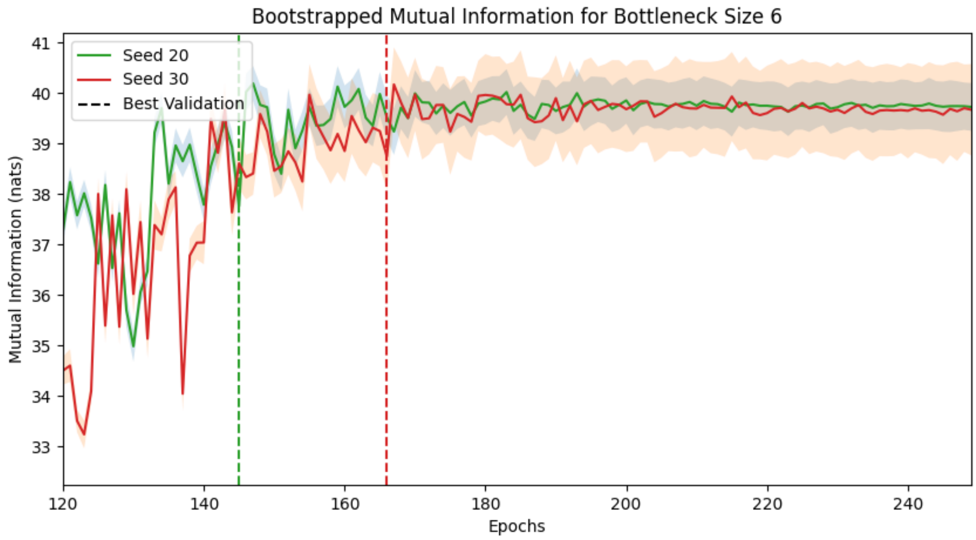


Figure 7: Mutual Information per Epoch for Bottleneck Dimension 4

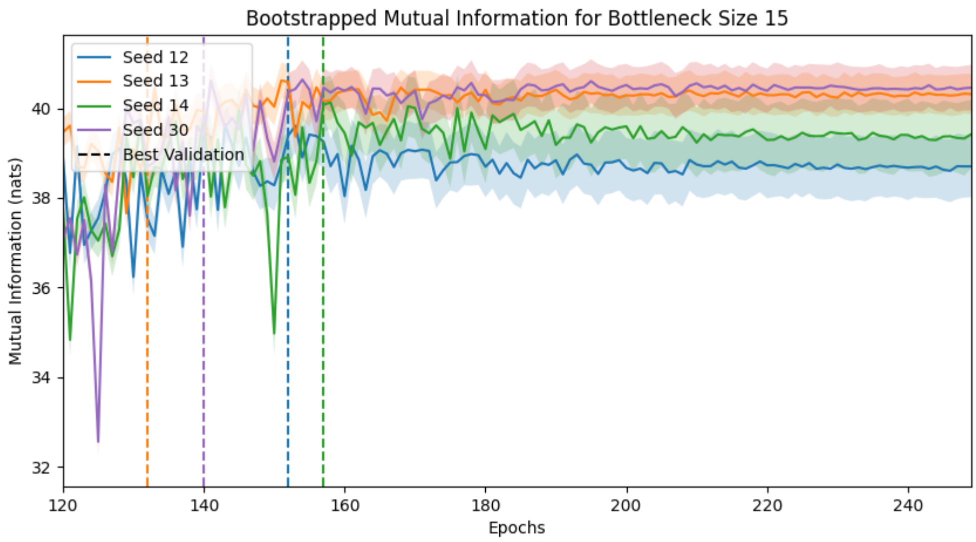


Figure 8: Mutual Information per Epoch for Bottleneck Dimension 4

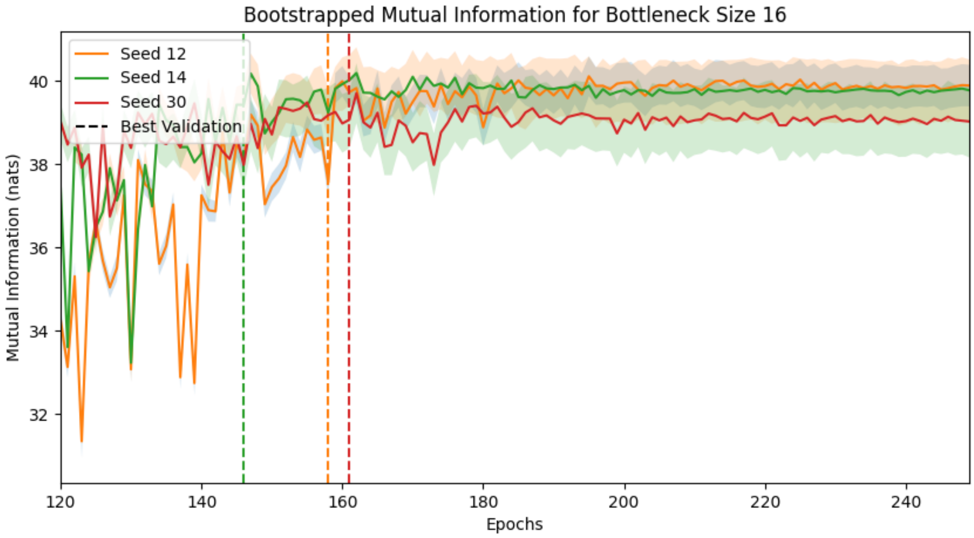


Figure 9: Mutual Information per Epoch for Bottleneck Dimension 4

These results can be described as follows: One does observe a general trend of increasing

Table 4: Final Esimation of CE per Bottleneck Dimension (all entropy values in nats)

Bottleneck	CE	Seed error	Model error	No. samples
4	38.81 ± 0.71	0.39	0.12	3
6	38.49 ± 0.95	0.83	0.06	4
15	39.90 ± 0.42	0.13	0.05	5
16	39.57 ± 0.46	0.13	0.09	4

mutual information for increased bottleneck dimension, though the increase is not monotonous. In all cases, the seed error is larger than the model error, indicating training instablity to be the largest source of error in this setup.

This indicates the seed error being overestimated? Maybe have to exclude some model. Talk about decision mechanisms and how this introduces bias

Results if including selection bias

9 Interpretation

Possible interpretations of the results:

The flow seems to be overcondident, because of the long negative tail in log probability plot. If choosing MI estimation from earlier epochs, get smaller error! This is due to observed overfitting and still improving performance.

Could overconfidence be due to the choice of priors?

Alternative interpretation:

The CNN learns the latent space. If the weights of the CNN aren't updated enough and the flow is doing too much of the heavy lifting, then the latent space structure might be hard to estimate, resulting in long negative tail.

Check to see wether the extreme models are the reason for this, or if this is a problem with the flow.

The seed uncertainty for this model is very high, given some numbers here. The high sensitivity towards weight initialization points towards unstable training.

Once possible explanation for this instability could be the choice of hyperparameters. For example, a large learning rate might magnify an initially chaotic training process, causing the model to end in different minimia of the loss landscape.

Another possibility is the choice of architecture. For example, the flow might be too deep given the data. While one might expect the need for a deep architecture to infer 15-dimensional labels, these labels are likely highly correlated across dimensions because of curve shape etc, making the problem easier than naively expected. The fact that one observes an early saturation in mutual information across bottleneck sizes supports the latter statement.

Of course, there is always the possibility of there being too little training data. Choosing larger parts of the training set for validation and testing indeed showed an increase in overfitting towards the end of training.

Something about consistency of MI estimate, since e.g. CE estimate is not larger zero from beginning. Maybe cite from paper?

From paper: "In par- ticular, higher estimated MI between observations and learned representations do not seem to indi- cate improved predictive performance when the representations are used for downstream supervised learning tasks (Tschannen et al., 2019)."

This is also reflected in these data, where MI is sometimes higher! In case not making mistake in indexing in results code.

10 Outlook

How one could incorporate information geometry for similar problems

Any idea how to numerically resolve F_{ij} for this flow?

Alternative choices than to model instead of $p(x|y)$ and $p(x)$?

A Network Implementation

Network architecture, training details, losses